ICT2122 · Object Oriented Programming

Worked Examples: Interfaces and Exceptions

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

An interface states behavior that multiple classes can implement. Exceptions report failures that cannot be represented as ordinary return values. Good object-oriented design makes ownership and error responsibility explicit.

2. Example and reasoning

A file importer can depend on a Repository interface. A fake repository supports deterministic tests; a JDBC repository performs persistence. A validation exception should identify invalid input without leaking SQL or filesystem details.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “worked examples: interfaces and exceptions” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.